



Precision Industries
Testing Department

GENERATING-SET LOAD TEST

Model No: PI 438 C	60 Hz
Type: Close	Job No: 8735
Color: Orange	Date: 03/12/2017

Serial Number:								
PI Model No:	PI 438 C	Prime Power:	400	[kVA]	P.F (Load):	1	Voltage:	380

Generating Set Details:

Engine:	Cummins	Model No: NTA855 - G3	Serial No: 25430956
Alternator:	Stamford	Model No: HCI444F1	Serial No: N17F240192
Controller:	DSE	Model No: 6120	Serial No: 6598807
C. Breaker:	Metasol- 800A	Model No: ABN803C	Serial No: N/A
C.Transformer:	Lovato 800/5A	Model No: N/A	Serial No: N/A
AVR:	Stamford	Model No: AS 440	Serial No: 111593 - 2599
Cooling System:	Water cool	Enclosure Type: Canopy	Protection Class: IP23

Load Test Ambient Conditions:

Temperature	27	[°C]	Altitude:	0	[m]	Humidity:	30	%
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Generating Set Rating:

Testing Full Load Current:	486.19	Amps	Nominal Current:	607.7	Amps
Power Correction Factor (De-rating):	0.00	%	Corrected Prime Power:	0.0	[kVA]

LOAD TEST

Load	Run Time	Freq	Speed	VOLTAGE L-L			CURRENT			Power	Coolant Temp	Oil Pressure	Charg-Alt.
				L1	L2	L3	L1	L2	L3				
%	min	Hz	RPM	V	V	V	A	A	A	kW	C	bar	VDC
0 %	5	60	1800	380	380	380	0	0	0	-	52	5.51	28.2
25 %	5	60	1800	380	380	380	124	122	122	81.0	76	4.89	28.2
50 %	5	60	1800	380	380	382	243	242	242	159.0	78	4.82	28.2
75 %	5	60	1800	379	380	382	369	368	369	244.0	81	4.61	28.2
100 %	10	60	1800	379	380	382	488	487	487	322.0	83	4.48	28.2
110 %	10	60	1800	379	380	382	531	530	530	349.0	86	4.41	28.2

LOAD ACCEPTANCE TEST

Suddenly Block Load %	Voltage [V]	Frequency [Hz]	Current [A]	Response Time [sec]
50%	380	60.0	243	0
0%	383 - 380	61.4 - 60.0	0	2
52%	377 - 380	58.8 - 60.0	246	2

PROTECTION SYSTEM TEST

Over Frequency SD	Under Frequency SD	Over Voltage SD	Under Voltage SD	Oil Pressure Warning/SD	Coolant Level Warning/SD	Coolant Temp Warning/SD
67	53	460	340	2.0/1.0	N/A	95 / 104 C

ACCESSORIES DETAILS

Model:		Serial No.	
Model:		Serial No.	

PI Document no.	PI-TST-LT-01	Rev.	00
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Test Certificate

FOR
SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATORMachine No. **N17F240192**

Type Test has been conducted on similar design machine
STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: HCI444F1	AVR: AS440	AVR NO: 1115932599	ENCLOSURE: IP23
RATED KVA: 400	RATED VOLTS: 400	RATED AMPS: 577.4	FREQUENCY: 50
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous
EXC. VOLTS: 48	EXC.AMPS: 2.3	AMB TEMP: 40	PHASE: 3
			WIRE: 4-Wire

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.01344	2+	OK	H
ROTOR	1.401	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	9.5	.52	0.00
50	400	577.4	400	.8	320	48	2.3	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator.HC4.4.F.1.311.,1,,14,.AS440.Black - RAL9011
 Remarks:

Date: 18-JUN-17	Job No: 3484497-001	AUTHORISED BY Rameshwar B Karkhande Quality Assurance
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Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **HCI444F1**Serial Number : **N17F240192**

Manufactured Date : 18-JUN-17

KVA: **400**Authorized By Quality Assurance: **Rameshwar B Karkhande**